Planner Graph

Is it possible to draw a graph in a plane without its edges crossing over?

S. S. Satapathy

Combinatorial vs Geometric Graphs

- A **graph G** exists as **an abstract object**, that can be **defined** as **G** = (***V, E,*** **Ψ**).
- For example, an abstract graph *G*1 can be defined as ***G*1** = (***V, E,*** **Ψ**),
  - where the **set *V*** consists of the five objects, ***V*** = {***a, b, c, d, e***},
  - and the **set *E*** consists of seven objects (none of which is in set *V*), ***E*** = {**1, 2, 3, 4, 5, 6, 7**},
  - and the **relationship between** the **two sets** is defined by the mapping **Ψ**, which consists of

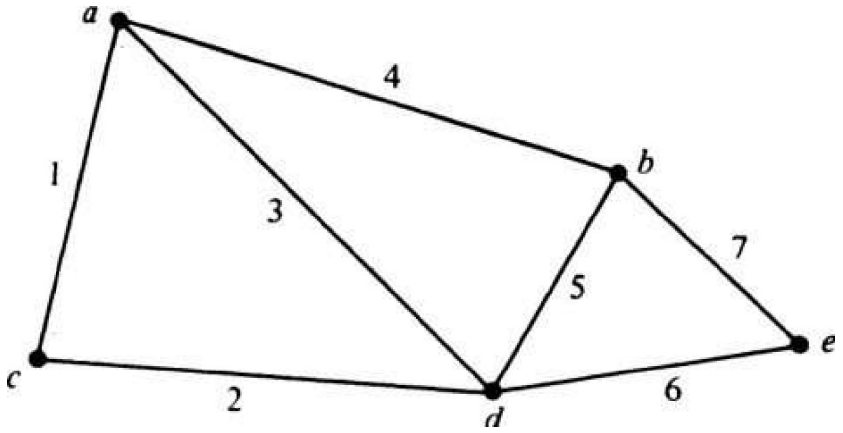
$$\Psi = \begin{bmatrix} 1 \longrightarrow (a, c) \\ 2 \longrightarrow (c, d) \\ 3 \longrightarrow (a, d) \\ 4 \longrightarrow (a, b). \\ 5 \longrightarrow (b, d) \\ 6 \longrightarrow (d, e) \\ 7 \longrightarrow (b, e) \end{bmatrix}$$

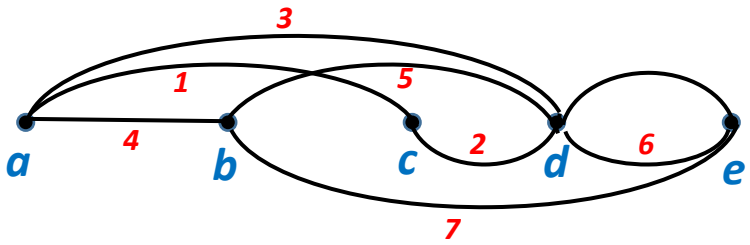
- This **graph G1** can be **represented** by **geometric figures** in **many different ways**.

Combinatorial vs Geometric Graphs

- The **graph G1** = (***V, E, Ψ***) where ***V*** = {***a, b, c, d, e***}, and ***E*** = {**1, 2, 3, 4, 5, 6, 7**}, and the **relationship between** the **two sets Ψ** defined as shown can be **represented** by **geometric figures** in **many different ways**, two examples are given below.

$$\Psi = \begin{cases} 1 \longrightarrow (a, c) \\ 2 \longrightarrow (c, d) \\ 3 \longrightarrow (a, d) \\ 4 \longrightarrow (a, b). \\ 5 \longrightarrow (b, d) \\ 6 \longrightarrow (d, e) \\ 7 \longrightarrow (b, e) \end{cases}$$





Combinatorial vs Geometric Graphs

- There are some **graphs** for those where there **exists some geometric representation on a plane** such that **no two of its edges intersect**.
- For some other graphs, we can't have any **geometric representation on a plane** such that **no two of its edges intersect**.

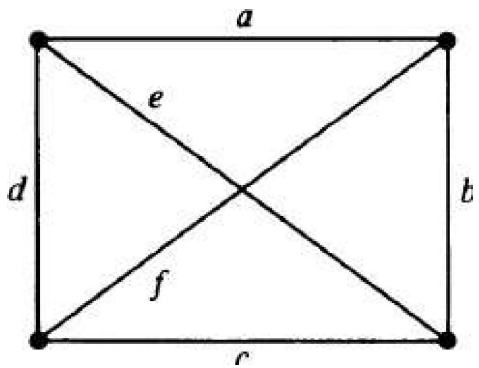


Fig. 1

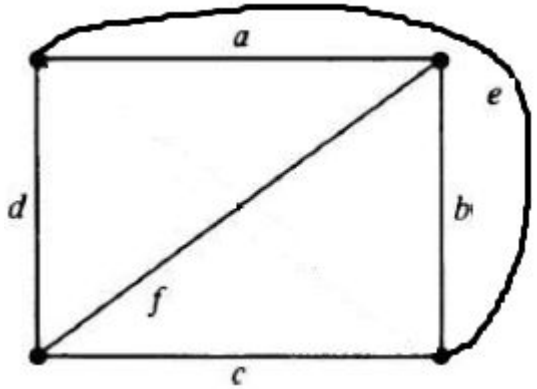
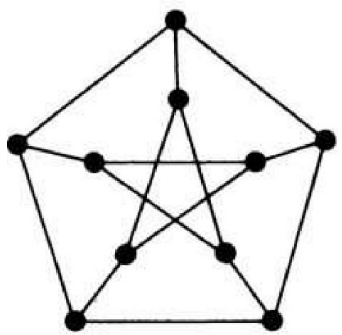
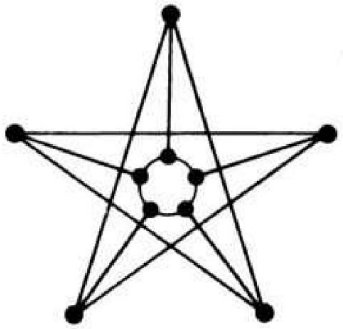
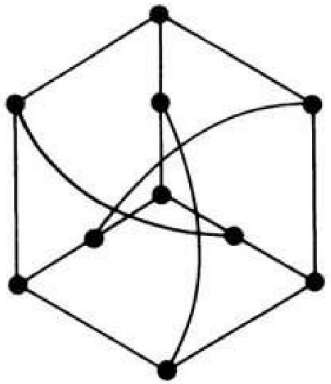


Fig. 2







Isomorphic graphs

Isomorphic graphs

Planar Graphs

- A **graph** ***G*** is said to *be* ***planar*** if there exists some geometric representation of *G* which can be drawn on a plane such that no two of its edges intersect, otherwise is called ***nonplanar***.
- The **edges e** and **f cross each other** in the graphical representation of a Graph in **Fig. 1**, whereas the edges **do not cross** in the representation of the same graph in **Fig. 2**.
- **Fig. 3** shows a **nonplanar graph** as there is **no graphical representation** of the graph **without crossing of edges**.
- **Planar graphs** can be used for **2-dimensional models** for certain **applications**, whereas **nonplanar graphs** require **higher dimensions**.

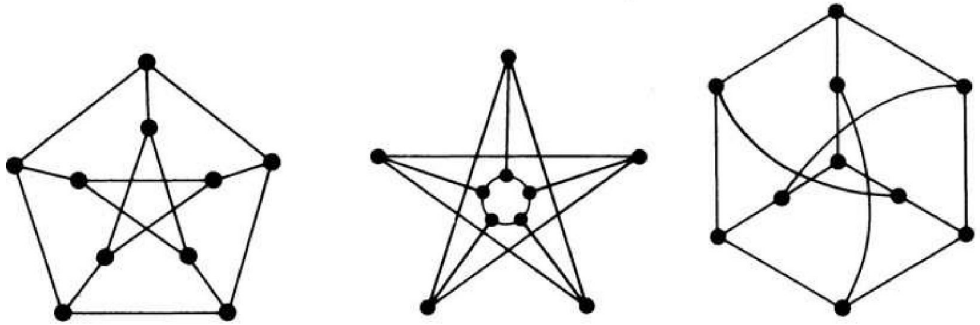


Fig. 3 ***nonplanar*** **graph**

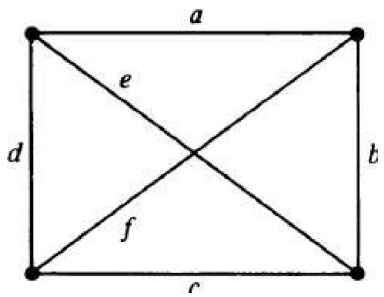


Fig. 1

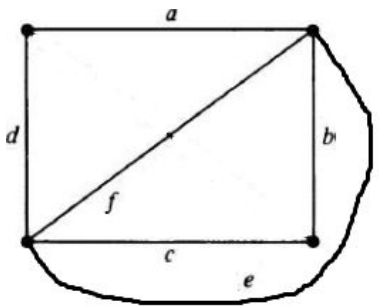


Fig. 2

Planar **graph**

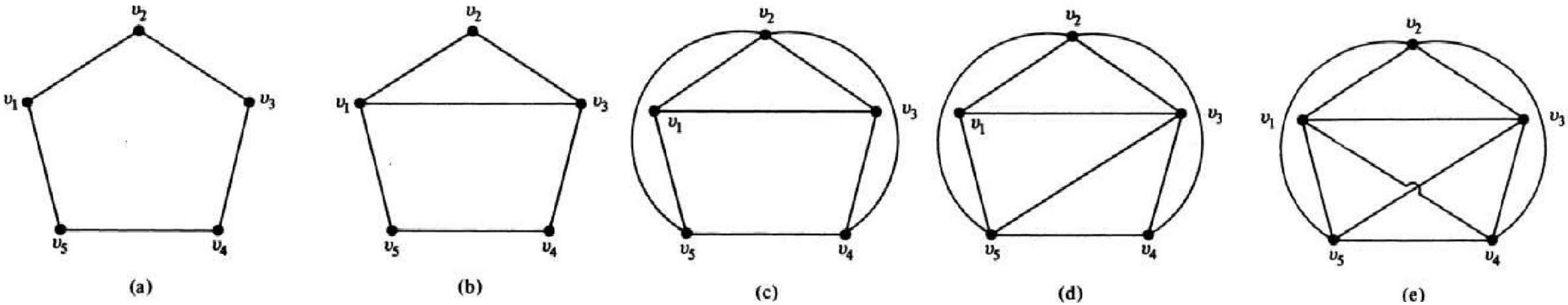
Kuratowski's graphs

- **How can we know if a graph *G*** [which may be given by **an abstract notation *G* = (*V*, *E*, Ψ)** or by one of its geometric representations] is **planar or nonplanar**?
- To **answer** this **question**, we need to know about **two fundamental nonplanar graphs**, called **Kuratowski's graphs**, after the **Polish mathematician Kasimir Kuratowski**, who discovered their unique property.
- **Graph 1**: complete graph of five vertices
- **Graph 2**: a regular connected graph with six vertices and nine edges

- **Note**: All the graphs with <=4 vertices are always planar.

Kuratowski's graph 1

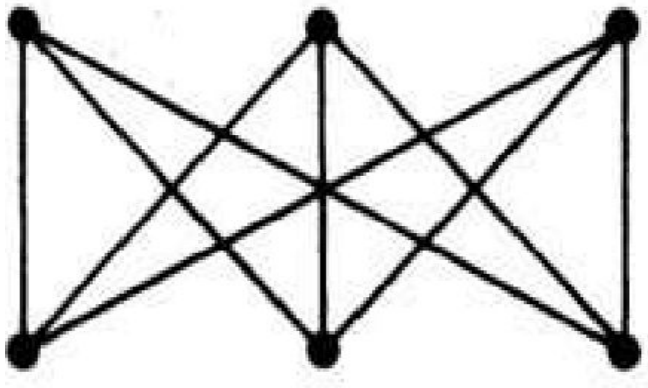
- **Graph 1**: complete graph of five vertices



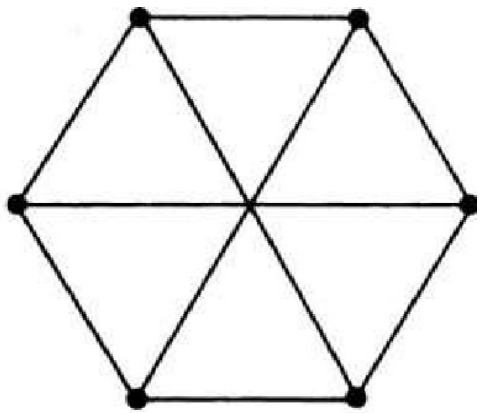
Stepwise placing edges to show that the graph can't be drawn without a crossover. Thus the graph cannot be embedded in a plane.

Kuratowski's graph 2

- **Graph 2**: a **regular** **connected graph** with **six** **vertices** and **nine** **edges**



(a)

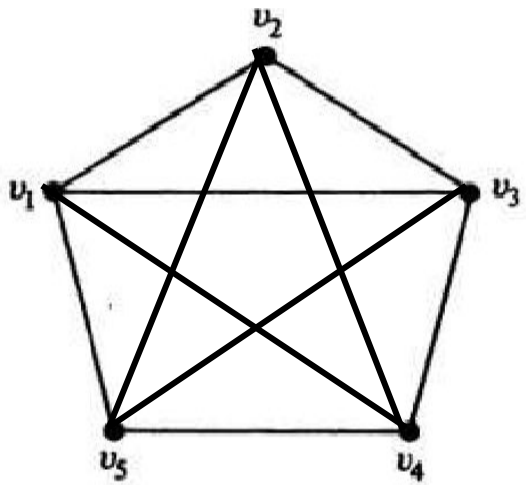


(b)

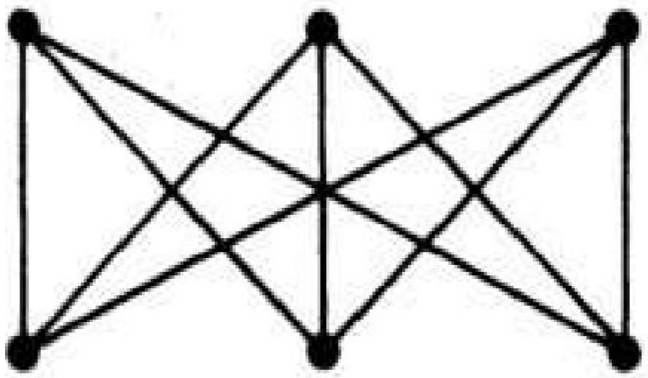
Two common geometric representations of Kuratowski's graph 2 showing its nonplanarity.

Common Properties of Kuratowski Graphs

- Both are **regular** graphs.
- Both are **nonplanar**.
- **Removal** of **one edge** or **a vertex makes** each a **planar** graph.
- **Kuratowski's first graph** is the **nonplanar graph** with the **smallest number** of **vertices**
- **Kuratowski's second graph** is the **nonplanar graph** with the **smallest number** of **edges**.
- Both are the **simplest nonplanar** graphs.
- **Kuratowski's first** graph is denoted by ***K*5** and the **second graph** by ***K*3.3**—letter *K* being for Kuratowski.



Kuratowski's graph 1



Kuratowski's graph 2

Straight-line Representation of a Graph

- **Theorem 1**: **Any simple planar graph** can be **embedded in a plane** such that **every edge** is **drawn** as a **straight line** segment.
- **Note**: it is **necessary** for the **graph to be simple** because a **self-loop** or one of two **parallel edges** **cannot be drawn** by a **straight line** segment.

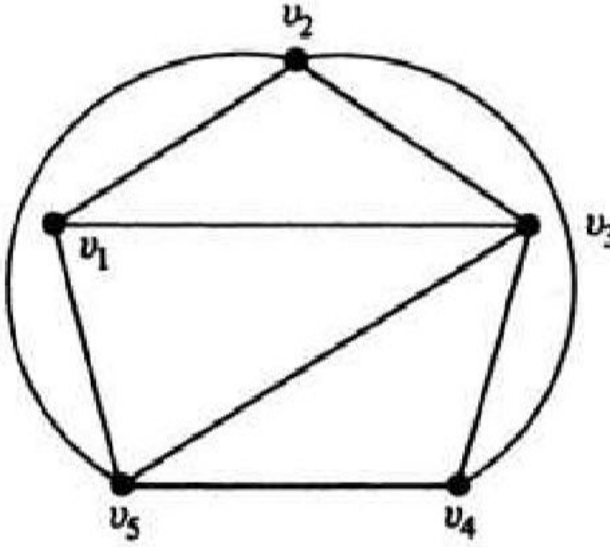
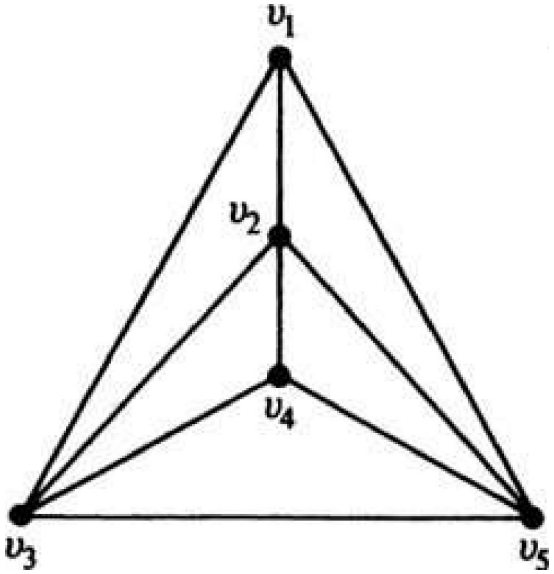


Fig. 1. A planar graph



Straight-line representation of the graph in Fig. 1.

Regions in a Planar Graph

- **Region:** A **plane representation** of a **graph divides** the **plane** into **regions** (also called windows, faces, or meshes), as shown in Fig.
- A **region** is **characterized** by the **set of edges** (or the set of vertices) **forming** its **boundary**.
- **Infinite Region**: The **portion** of the **plane lying outside a graph** embedded in a plane, such as **region 4** in Fig., **is infinite in extent**. **Such a region** is **called** the **infinite**, **unbounded**, **outer**, or **exterior region** for **that particular plane representation**.

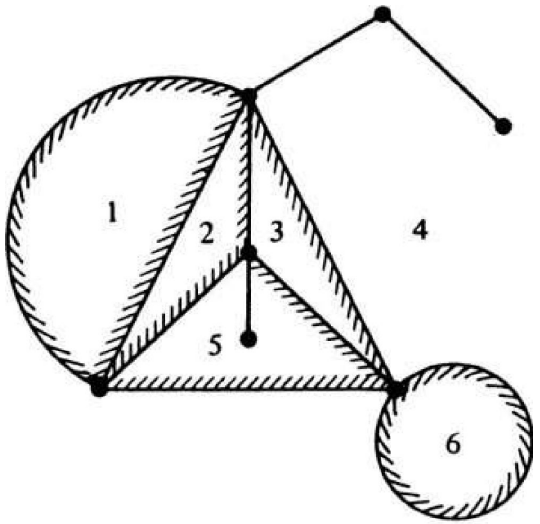


Fig. Plane representation (the numbers stand for regions)

Regions in a Planar Graph

- **Region** is a **property of the specific plane representation** of a graph and **not of an abstract graph** per se.
- By changing the embedding of a **given planar graph**, the **infinite region can be changed**. For instance, Fig. a and b are two different embeddings of the same graph. The **finite region v1 v3 v5** in **Fig. a becomes** the **infinite region in Fig. b**. In fact, any region can be made an infinite region by proper embedding.
- Though **a planar graph may have different plane representations**, the **number of regions** resulting from **each embedding** is the **same**. Graphs **in both Fig a**, and **Fig b**, **have 6 regions** each.

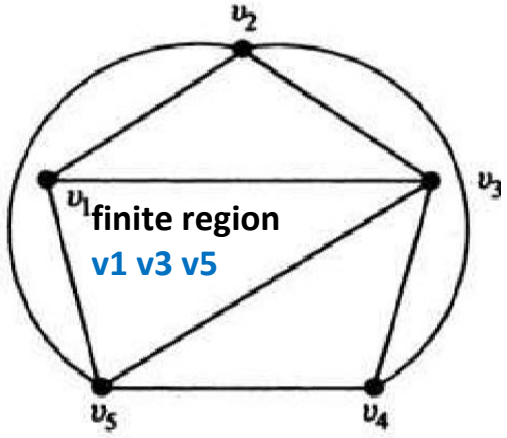


Fig. a

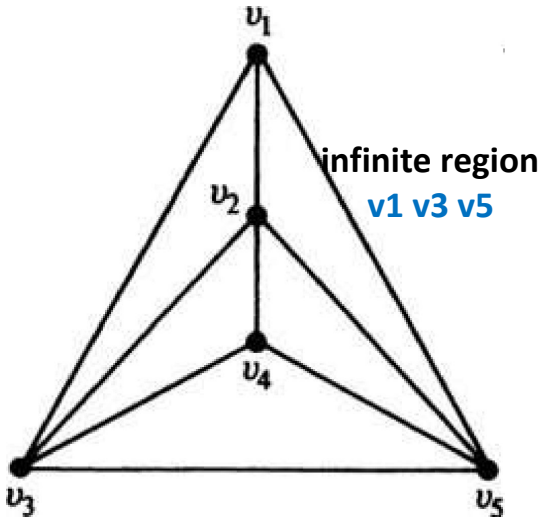
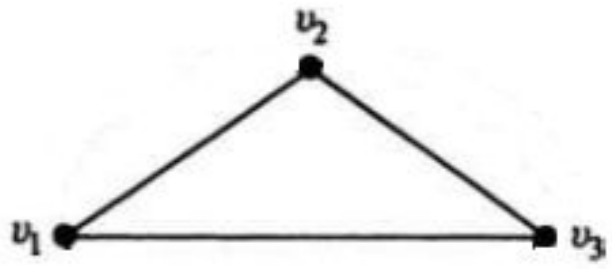


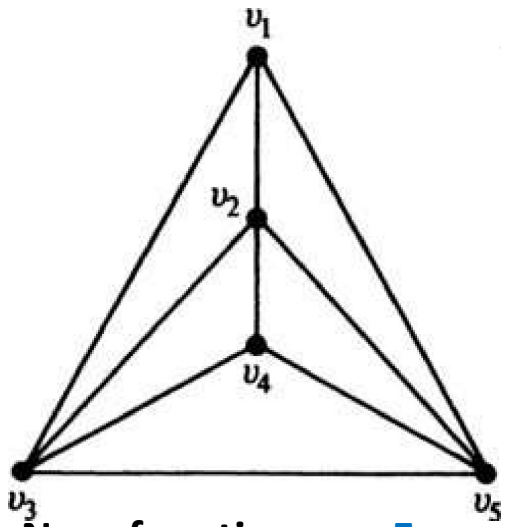
Fig. b

Number of Regions (Euler's Formula)

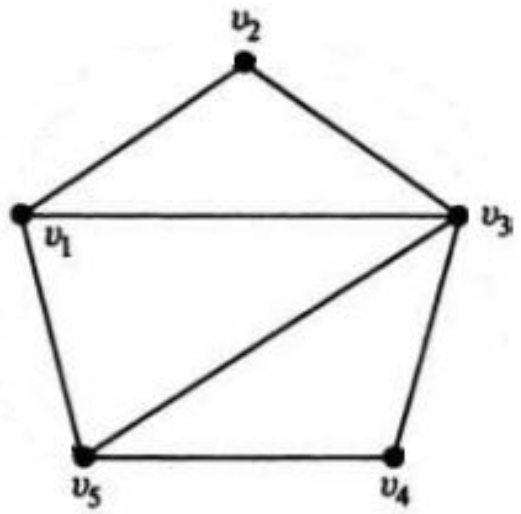
- For a **connected planar graph** with **n vertices** and **e edges**, **no. of regions f = e − n + 2** (**Euler's Formula**)



No. of vertices n = 3
No. of edges e = 3
No. of regions f = 2
e − n + 2 = 3 − 3 + 2 = 2



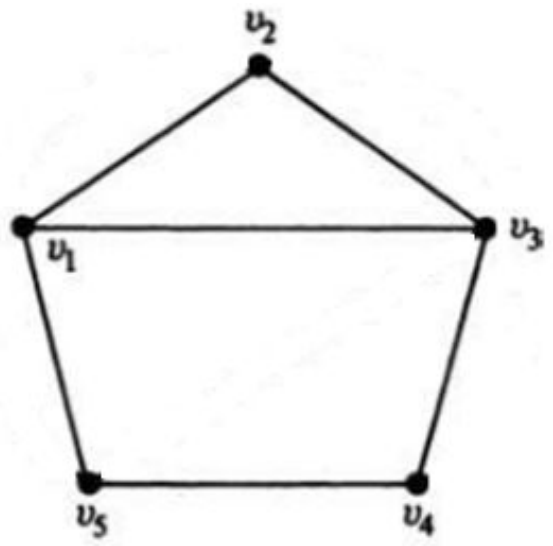
No. of vertices n = 5
No. of edges e = 9
No. of regions f = 6
e − n + 2 = 9 − 5 + 2 = 6



No. of vertices n = 5
No. of edges e = 7
No. of regions f = 4
e − n + 2 = 7 − 5 + 2 = 4

Number of Regions in a Planar Graph

- **Inequality 1:** In any **simple, connected planar graph** with **f regions**, **n vertices**, and **e edges** (**e > 2**), always the **inequality e >= (3/2)f holds**
- **Proof**:
  - Since each region is bounded by at least three edges and each edge belongs to exactly two regions, we have **2e >= 3f** or **e >= (3/2)f**



No. of vertices n = 5
No. of edges e = 6
No. of regions f = 3
e (=6) and
(3/2)f=(3/2)3=4.5

Condition for Planarity of a Graph

- **Inequality 2:** In any **simple, connected planar graph** with **f regions**, **n vertices**, and **e edges** (**e > 2**), always the **inequality e <= 3n – 6 holds**
- **Proof**:
  - **e >= (3/2)f** [From the 1st inequality, we have
  - **e >= (3/2)(e-n+2)** [Considering Euler's formula f=e-n+2
  - **2e >= 3(e-n+2) = 3e – 3n + 6**
  - **e <= 3n – 6**
- **This inequality** is **useful** in **finding out if a graph is nonplanar**. For example, the **complete graph** of **5 vertices** is **nonplanar** because **n=5**, **e=10** but **3n – 6 = 9 < e**
- **Note:** This inequality (**e <= 3n – 6**) is only a **necessary, but not a sufficient condition** for the **planarity of a graph**, i.e. **every simple planar graph must satisfy this inequality**. **But satisfaction of this inequality does not guarantee** the **planarity** of a graph. E.g. **Kuratowski's second graph**, **e = 9**, **3n – 6 = 3*6 – 6 = 12 but the graph is nonplanar**.

Geometric Dual of a Graph

- In the **graph G** of **Fig. a**, **place six points** $p_1, p_2, \ldots, p_6$, **one in** each of the **regions** or faces $F_1, F_2, \ldots, F6$, as shown in **Fig. b**.
- **Join** these **six** points $p_1, p_2, \ldots, p_6$ according to the following procedure:
  - If two regions F_i and F_j are **adjacent** (i.e., have a common edge), **draw** a **line joining** points p_i and p_j **that intersects the common edge** between F_i and F_j exactly once.
  - If there is **more than one edge common** between F_i and F_j, **draw** one **line** between points p_i and p_j **for each** of the **common edges**.
  - For **an edge e lying entirely in one region**, say F_k, **draw** a **self-loop** at point p_k **intersecting e** exactly once.
- The **new graph G*** (in **red color Fig. b**) consisting of six

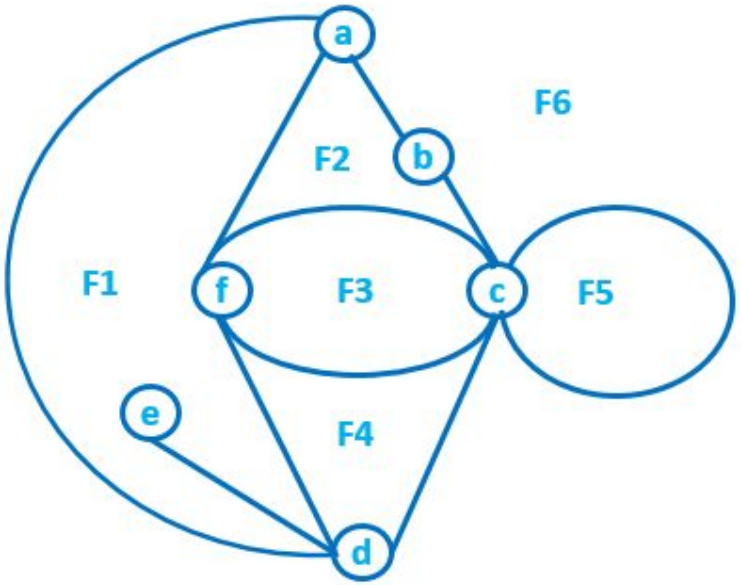


Fig. a. Graph G
No. of vertices n = 6; No. of edges e = 10; No. of regions f = 6

Geometric Dual of a Graph

- **draw** a **line joining** points p_i and p_j **that intersect the common edge** between F_i and F_j exactly once.
- If there is **more than one edge common** between F_i and F_j, **draw** one **line** between points p_i and p_j **for each** of the **common edges**.
- For **an edge e lying entirely in one region, draw** a **self-loop** at point p_k **intersecting** e

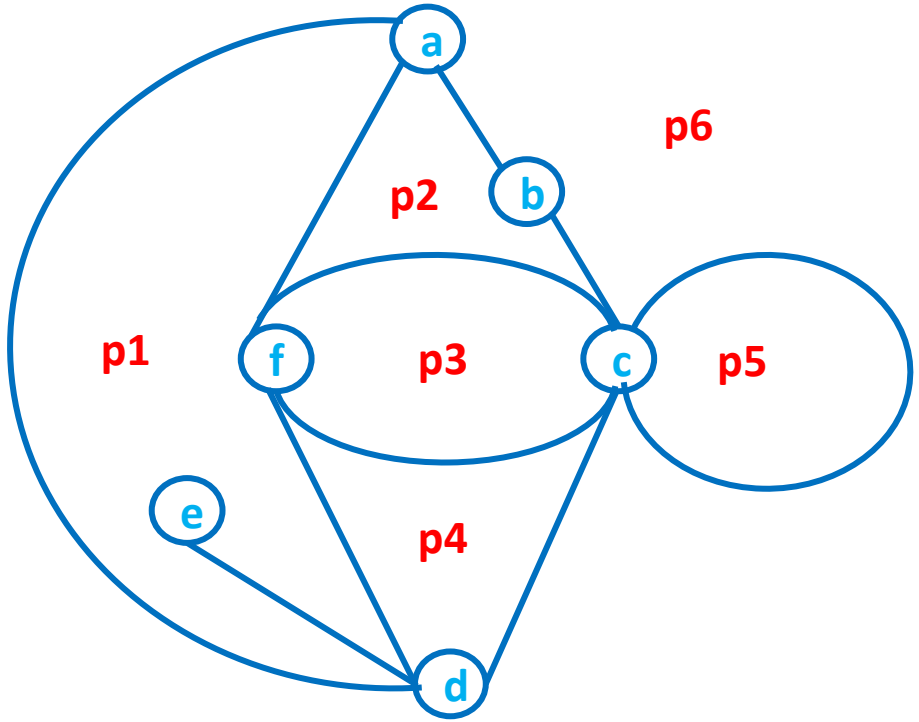


Fig. a. Graph G
No. of vertices n = 6; No. of edges e = 10;
No. of regions f = 6

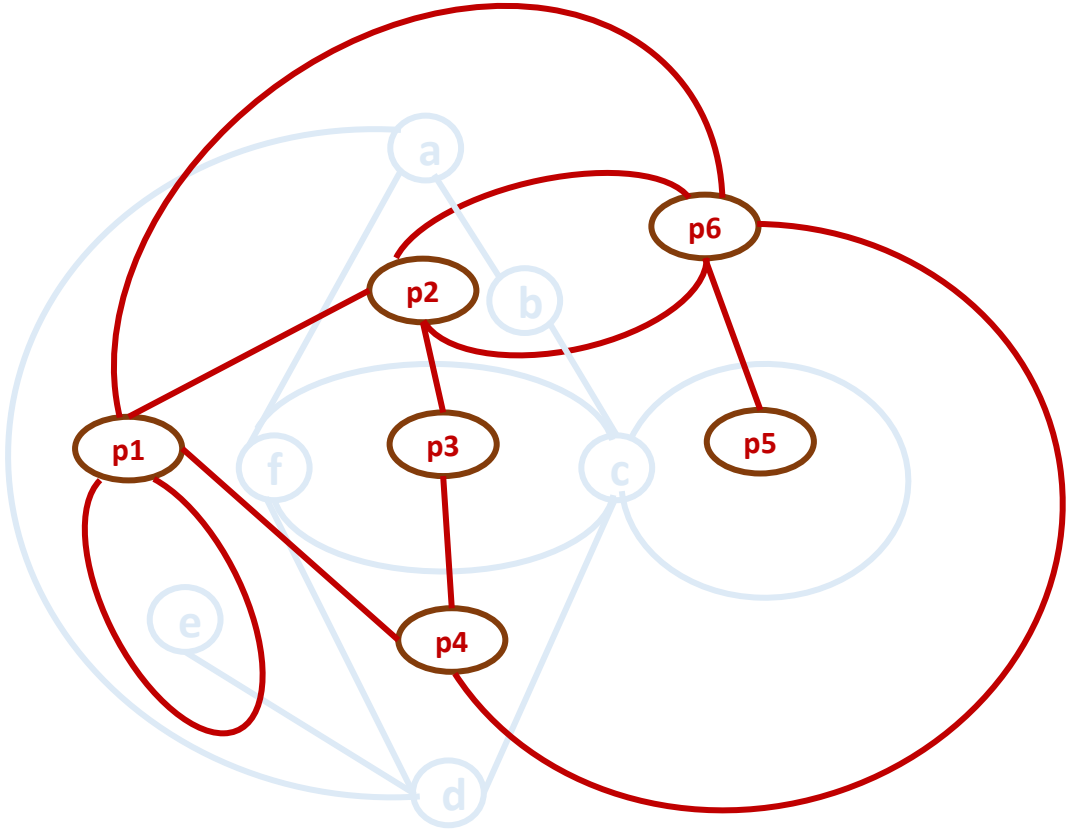
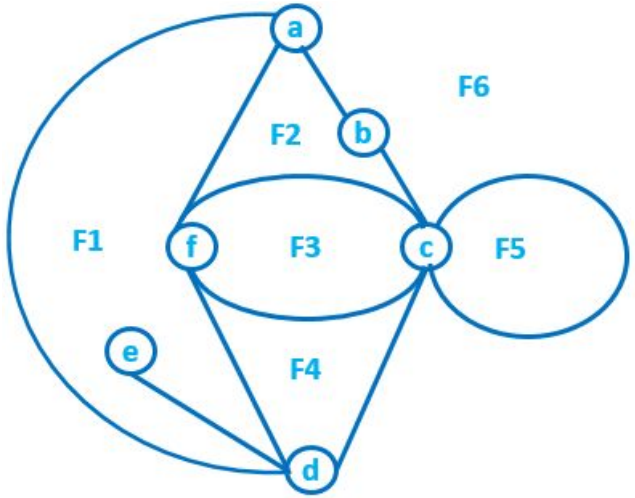


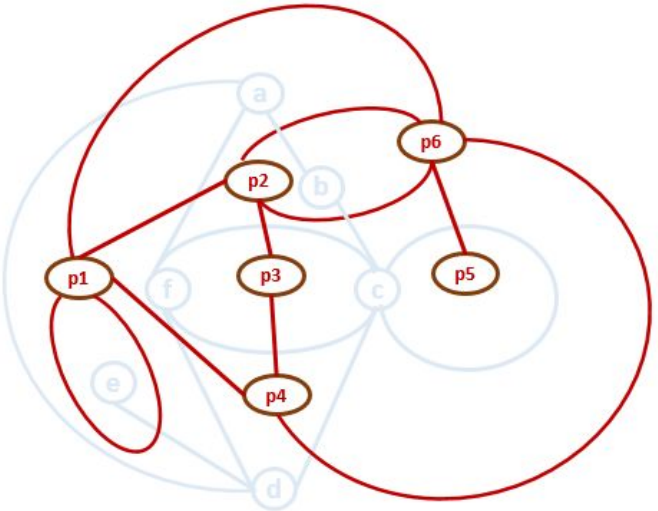
Fig. b. Graph G*
No. of vertices n = 6; No. of edges e = 10;
No. of regions f = 6

Geometric Dual of a Graph

- There exists **one-to-one correspondence between** the **edges of graph G** and its **dual G*– one edge of G* intersecting one edge of G**.
- Some simple **observations** on the **relationship between** a **planar graph G** and its **dual G*** are:
  - An edge forming a **self-loop in *G*** yields a **pendant edge** in ***G****.
  - A **pendant edge** in ***G*** yields a **self-loop** in ***G****.
  - **Edges that are in series** in ***G*** produce **parallel edges** in ***G****.
  - **Parallel edges** in ***G*** produce **edges** in **series** in ***G****.
  - The **number of edges constituting the boundary** of a **region F_i** in ***G*** is **equal to** the **degree of the corresponding vertex p_i** in ***G****, and **vice versa**.
  - **Graph G*** is also embedded in the plane and is therefore **planar**.
  - **G*** and **G** are **dual of each other**. Therefore, **G** and **G*** are **called dual graphs**.



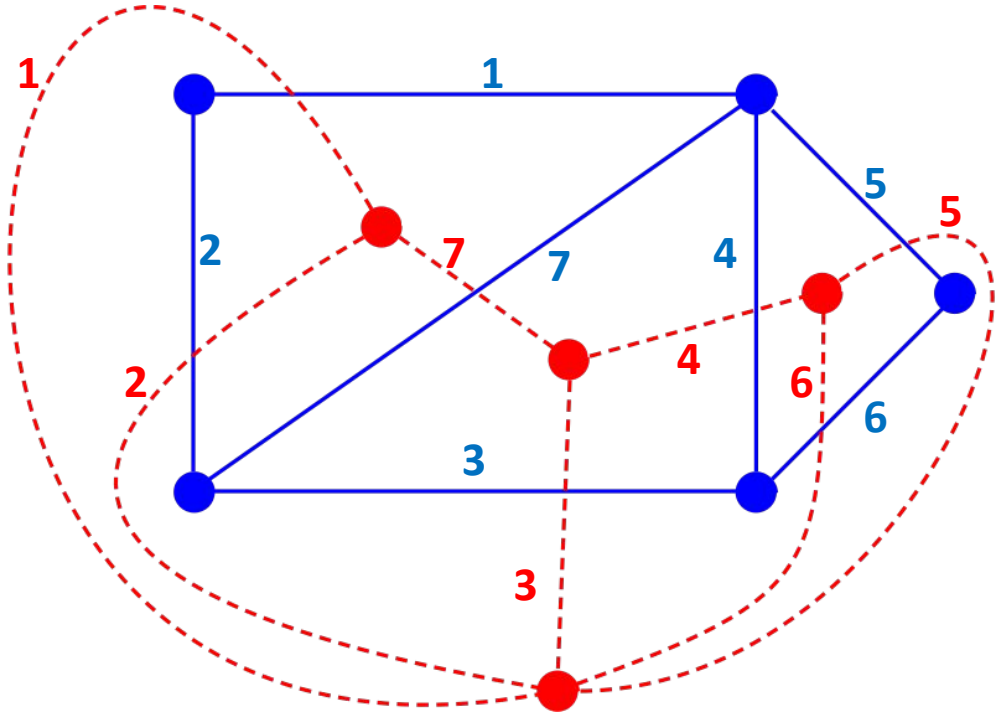
Graph G



Dual graph G*

Combinatorial Dual of a Graph

- A **necessary** **and** **sufficient** **condition** for **two planar graphs** **G1** and **G2** to be **duals** **of each other** is as follows:
  - There is a **one-to-one** **correspondence between** the **edges** in **G1** and the **edges** in **G2** such that a **set of edges** in **G1** forms a **circuit** **if and only if** the **corresponding set** in **G2** forms a **cut-set**.

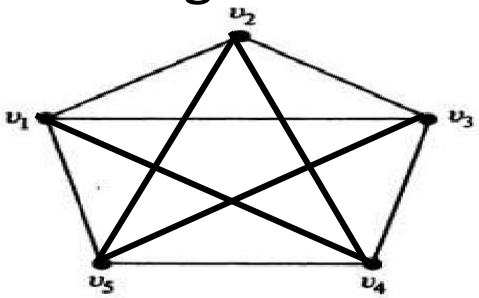


{1,7,3} is a cut-set in G1
but
{1,7,3} is a circuit in G2

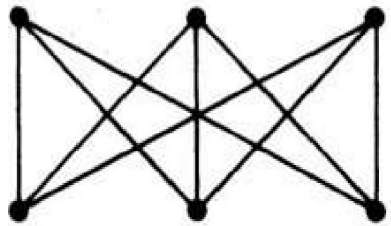
{4,7,3} is a circuit in G1
but
{4,7,3} is a cutset in G2

Thickness and Crossings of a Graph

- **What** is the **minimum number of planes** necessary for representing a graph?
  - The **least number of planar subgraphs** **whose union** is the **given graph G** is called the **thickness** of **G**.
  - The **thickness** of a **planar graph** is **one**. The **thickness** of each of **Kuratowski's graphs** is **two**. The **thickness** of an **arbitrary graph** is in general, **difficult to determine**.
- What is the **fewest number of crossings** (or **intersections**) necessary in order to **draw** the **graph in a plane**?
  - The **crossing number** of a **planar graph** is **zero**
  - The **crossing numbers** of either of **Kuratowski's graphs** is **one**
  - No formula exists to give the crossing number of an arbitrary graph.



Kuratowski's graph 1



Kuratowski's graph 2